Lee 2020 CRC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

14/15 samples (≥151 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); both lobes (Embryonic 6,147 / Cycling 4,823) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

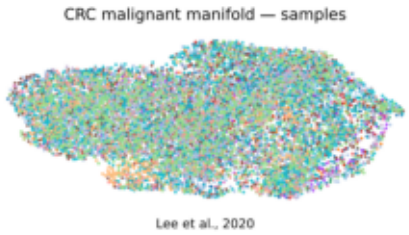
Gast/pSom co-vary (r=+0.49) and decline along pseudotime (ρ=−0.76/−0.56); Crest anti-correlated (r=−0.81/−0.83), rising to a terminal readout (ρ=+0.77; crest-high decile mean pt=0.95).

Gate 3 — hub mutual exclusivity → NA

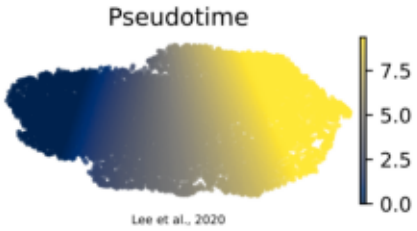
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

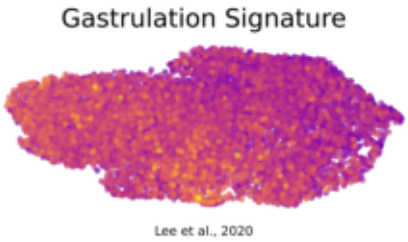
Manifold — samples



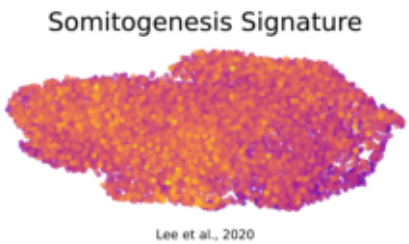
Pseudotime



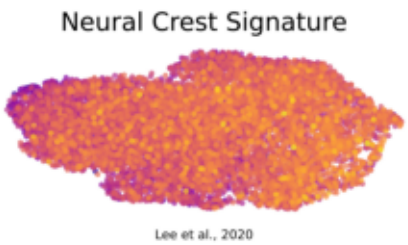
Gastrulation



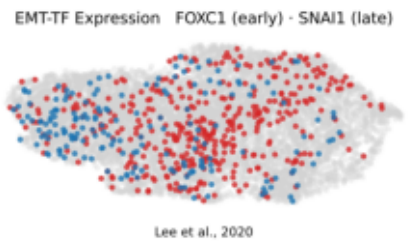
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Colorectal Carcinoma (Lee et al., 2020), 10x — 15 patients / 21,657 cells (master scope); malignant only: 11,075 load, 10,970 across 14 samples (≥151) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Lee's colorectal malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Unlike Kürten, the developmental axis spans both lobes (Crest terminus inside the cycling lobe), so the whole manifold is kept (Bischoff/Sun no-scoping precedent). EMT-TFs FOXC1 (early/root) → SNAI1 (late/terminus); ZEB1/TWIST near the detection floor (0.8%/0.2%); EMT-TF end-separation weak (Δpt=0.22).

Parameter choices

Min cells/sample	≥151 (keeps 14/15; gap 219→105)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Embryonic 6,147 / Cycling 4,823